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dms-viz: Structure-informed visualizations for deep mutational scanning and other mutation-based datasets

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Summary and Purpose

Understanding how mutations impact a protein's functions is valuable for addressing many biological questions. Deep-mutational scanning (DMS) offers an approach to characterize the impact of a huge number of mutations in parallel (Fowler & Fields, 2014). The wide application of DMS has greatly expanded the number of mutation-function datasets (Fowler et al., 2023). For instance, DMS has been used to determine how mutations to viral proteins affect antibody escape (Dadonaite et al., 2023), receptor affinity (Starr et al., 2020), and essential functions such as viral genome transcription and replication (Li et al., 2023). With the growth of sequence databases, in some cases the effects of mutations can also be inferred from phylogenies of natural sequences (Bloom & Neher, 2023) (Figure 1).

The mutation-based data generated by these approaches is often best understood in the context of a protein's 3D structure; for instance, to assess questions like how mutations that affect antibody escape relate to the physical antibody binding epitope on the protein. However, current approaches for visualizing mutation data in the context of a protein's structure are often cumbersome and require multiple steps and softwares. To streamline the visualization of mutation-associated data in the context of a protein structure, we developed a web-based tool, `dms-viz`. With `dms-viz`, users can straightforwardly visualize mutation-based data such as those from DMS experiments in the context of a 3D protein model in an interactive format. Visit https://dms-viz.github.io/ to use `dms-viz`.

Statement of Need

We wanted `dms-viz` to provide the following functionalities:

1. **Provide structural context**: The main objective of `dms-viz` is to simplify the process of visualizing mutation data with structural context by superimposing mutation measurements on a 3D protein structure. Additionally, it provides extensive control over the visual representation of the 3D structure.

2. **Accommodate diverse data types**: Although analyzing DMS data is a key goal of `dms-viz`, there are many types of mutation data. The tool can handle diverse data types via a command line interface that simplifies the process of converting data into a common format for analysis.

3. **Display multiple conditions**: With `dms-viz`, multiple experimental conditions can be

visualized concurrently, facilitating comparisons. Researchers can, for instance, easily visualize deconvolved antibody binding footprints from polyclonal sera (Yu et al., 2022).

4. **Maximize customizability**: Every dataset has specific needs for visual representation. Recognizing this, `dms-viz` offers a high level of customizability. Users can tailor filters, which are important for navigating large and possibly noisy datasets, and tooltips, ensuring that the nuances of their data are clear.

5. **Create compact interactive visualizations**: Interactive visualizations promote effective communication. `dms-viz` creates compact plots that can be incorporated into HTML presentation slides (e.g., https://slides.com/).

6. **Share findings with ease**: Users of `dms-viz` can generate shareable URL links for a customized visualization view. They can also save and share the `JSON` specification files created by the command line interface, ensuring that data can be accessed easily.

7. **Preserve data privacy**: `dms-viz` allows users to visualize proprietary structures and analyze sensitive data in their browser without uploading their datasets to a remote server or storing them in a public repository.

Our group previously created a tool called `dms-view` (Hilton et al., 2020) that has some of the functionalities listed above. However, we designed `dms-viz` to be more customizable and comprehensive to handle a wider diversity of experimental designs and questions.

Design and Usage

Using `dms-viz` involves three components. First, using the command line tool `configure-dms-viz`, available as a Python package on PyPI (https://pypi.org/project/configure-dms-viz/), the user formats their data into a `JSON` specification file (see the documentation for details on the `JSON` schema). Then, the user uploads this specification file to `dms-viz.github.io`, a web-based interface written in Javascript, `D3.js`, and `NGL.js` (Rose et al., 2018). Finally, the specification file can either be shared directly or hosted remotely to generate a shareable URL link (Figure 2).

Upon uploading the specification file to `dms-viz`, users will see a visualization composed of four components, as illustrated in Figure 3.

1. ***Context* plot**: Located at the top of the visualization, this component allows users to zoom into specific sites on the *Focus* plot while maintaining an overview of the entire dataset.

2. ***Focus* plot**: This plot shows a summarized view of the user's data. Every measured protein site is represented as a point providing a summary statistic of the effects of mutations at that site, and adjacent sites are connected with lines.

3. ***Detail* heatmap**: If the user is interested in the measurements for every mutation at a site, they can click on that site in the *Focus* plot. This will populate a heatmap with each individual mutation measurement at that site.

4. ***Interactive* structure**: When the user wants structural context for a given set of sites, they can drag a brush over the corresponding points in the *Focus* plot. This action will highlight those sites on an interactive 3D protein model.

To ensure the visualization remains compact, all configuration options are tucked away in a collapsible sidebar. See the documentation at https://dms-viz.github.io/dms-viz-docs/ for more information about how to use `dms-viz` along with detailed tutorials and examples.

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Examples

For additional scientific background and a walkthrough of the code that generates these visualizations, visit the documentation.

1. Mapping the neutralization profile of antibodies and sera against HIV envelope

Radford et al. (2023) mapped mutations to HIV envelope (Env) that affect neutralization by polyclonal human serum using a pseudotyping-based deep mutational scanning platform (Radford et al., 2023). See how `dms-viz` can be used to interactively visualize datasets with multiple antibody footprints on a single summary plot here.

2. Using mutation-fitness data to augment structure-guided drug design

Bloom and Neher estimated the fitness effects of mutations to all SARS-CoV-2 proteins by analyzing millions of human SARS-CoV-2 sequences (Bloom & Neher, 2023). See how `dms-viz` can be used to enhance structure-guided drug design by merging this data with structural views of a viral target like the SARS-CoV-2 main protease (Mpro) in complex with a bound ligand such as MAT-POS-e194df51-1 from the COVID Moonshot project (Boby et al., 2023) here.

3. Exploring the evolutionary potential of the influenza A polymerase PB1 subunit

Li et al. (2023) measured the effects of thousands of mutations to the PB1 subunit of the influenza RNA-dependent RNA polymerase on the replicative fitness of the lab-adapted influenza strain A/WSN/1933(H1N1) (Li et al., 2023). See how `dms-viz` can provide this dataset as an interactive resource here.

4. Visualizing the pathogenicity of genetic variants of an important tumor suppressor

Matreyek et al. (2018) used variant abundance by massively parallel sequencing (VAMP-seq) to characterize the effect of thousands of mutations on the intracellular abundance of PTEN, a tumor suppressor that is inactivated in many cancers (Matreyek et al., 2018). See how `dms-viz` can be used to identify clinically relevant mutations in human proteins here.

Conclusion

`dms-viz` is a valuable addition to the suite of computational tools available for analyzing, sharing, and visualizing mutation-based datasets, which includes MaveDB, ProtVar, and many others (Esposito et al., 2019). We designed `dms-viz` as a practical and user-friendly approach to visualizing mutation-associated data in the context of protein structures. Because `dms-viz` is capable of handling various data types and has options for both sharing and privacy, it should be applicable to visualization of a wide range of datasets.

Code Availability

- `dms-viz` is available at https://dms-viz.github.io/
- The documentation and information about developing `dms-viz` is available at https://dms-viz.github.io/dms-viz-docs/

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- The source code for `dms-viz.github.io` is available at https://github.com/dms-viz/dms-viz.github.io
- The source code for `configure-dms-viz` is available at https://github.com/dms-viz/configure_dms_viz

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Disclosures

JDB is on the scientific advisory boards of Apriori Bio, Aerium Therapeutics, Invivyd, and the Vaccine Company. JDB receives royalty payments as an inventor on Fred Hutch licensed patents related to deep mutational scanning of viral proteins.

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Figures

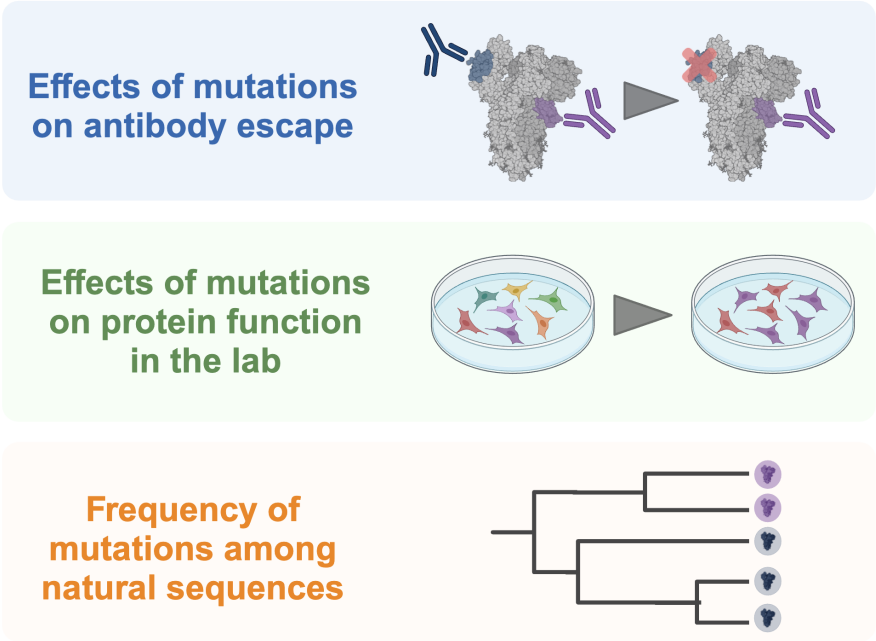

Figure 1: Large mutation-associated datasets are used in a variety of experimental contexts. They can be used to map antibody footprints on viral glycoproteins, assess the impact of mutations on protein function in a laboratory setting, and identify patterns of selection from natural mutation frequencies.

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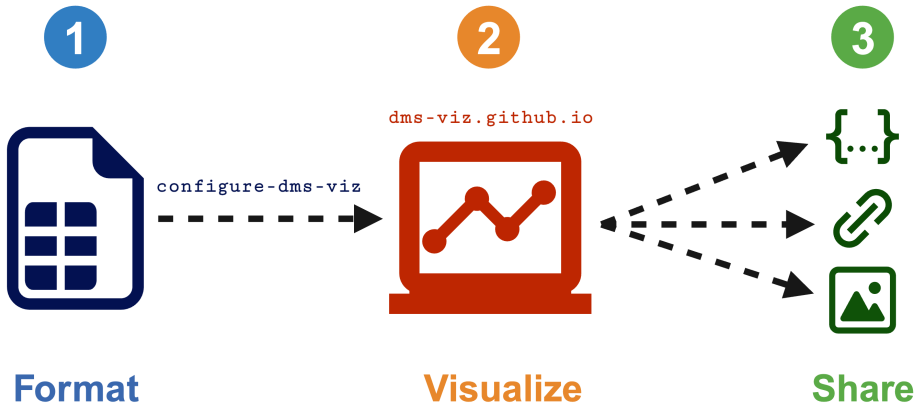

Figure 2: Using `dms-viz` involves three components. (1) The user formats their data using the command line tool `configure-dms-viz`. (2) The user takes the resulting `JSON` specification file and uploads it to `dms-viz.github.io`. (3) The user can choose to either share the `JSON` file, host the `JSON` file and generate a shareable URL link, or export static images.

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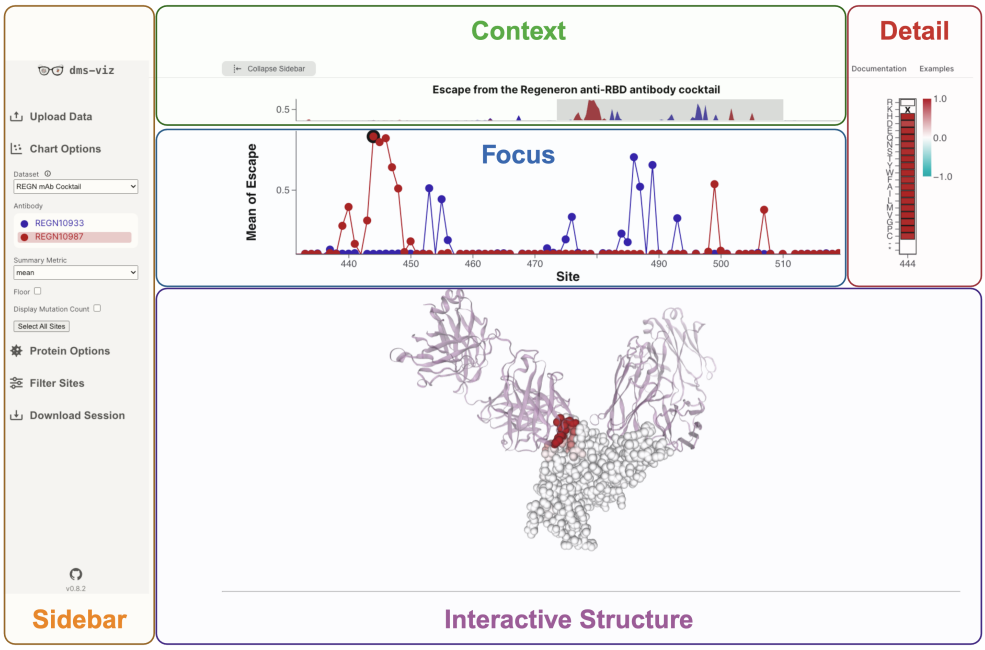

Figure 3: `dms-viz` provides a compact interface for exploring mutation-associated data. The visual component of `dms-viz` contains a line/point plot that shows a summary of the mutation-metric at all sites, in this case, mutation-escape from the constituents of a therapeutic antibody cocktail measured by DMS of the SARS-CoV-2 receptor binding domain (RBD) (Starr et al., 2021). The user can zoom into specific regions of interest while maintaining context of the whole dataset using the context plot. Additionally, users can click on points in the focus plot to get details on every mutation for each site in the detail plot. Finally, sites that are selected on the focus plot by dragging are shown on the interactive protein structure colored by the summary statistic. In this example, the structure shown is the SARS-CoV-2 RBD bound to both antibodies in the therapeutic cocktail (PDB: 6XDG). A collapsible sidebar is used to configure the visualization and select the condition on the interactive protein structure. By collapsing out of view, the sidebar makes the visualization an optimal size for integrating into online platforms like websites and HTML presentation slides.

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